

LECTURE 10: EXPECTATION AND VARIANCE

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PLAN

Expectation

1. Discrete and continuous random variables
2. Functions and linearity

Variance

1. Definition and properties
2. Sums and sample averages

Potential outcomes

1. Treatment effects and causal estimands
2. The fundamental problem of causal inference



EXPECTATION OF A DISCRETE RANDOM VARIABLE

What is the average value shown by a fair six-sided die?

y	1	2	3	4	5	6
$\Pr(Y = y)$	1/6	1/6	1/6	1/6	1/6	1/6

We multiply each possible value by the probability that it occurs:

$$E(Y) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = 3.5.$$

In general, if Y is a discrete random variable,

$$\mathbf{E(Y)} = \sum_y y \Pr(Y = y).$$

WHAT DOES EXPECTATION MEAN?

The expectation is a weighted average of all possible values of the random variable. Values that are more likely receive more weight.

If we independently draw $Y_1, \dots, Y_n$ from the same distribution, then, under mild conditions,

$$\frac{Y_1 + \dots + Y_n}{n} \longrightarrow E(Y).$$

This is the **law of large numbers**. It says that the sample average approaches the expected value as the number of draws becomes large.

Notice: the expectation does not have to be a possible value. A die never shows 3.5.

EXPECTATION OF A CONTINUOUS RANDOM VARIABLE

Suppose that Y is continuous with density f . We can't add the possible values: there are infinitely many, and each one has probability zero.

Instead, we integrate:

$$\mathbf{E}(\mathbf{Y}) = \int_{-\infty}^{+\infty} y f(y) dy.$$

Example. If $Y \sim \mathcal{U}[0, 2]$, then $f(y) = 1/2$ for $y \in [0, 2]$. Therefore,

$$\mathbf{E}(Y) = \int_0^2 y \frac{1}{2} dy = \left[\frac{y^2}{4} \right]_0^2 = 1.$$

This makes sense: the uniform distribution on $[0, 2]$ is symmetric around 1.

EXPECTATION OF A FUNCTION

If Y is a random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a function, then $g(Y)$ is also a random variable.

If Y is discrete,

$$\mathbf{E}[g(\mathbf{Y})] = \sum_y g(y) \Pr(Y = y).$$

If Y is continuous with density f ,

$$\mathbf{E}[g(\mathbf{Y})] = \int_{-\infty}^{+\infty} g(y) f(y) dy.$$

We don't need to find the distribution of $g(Y)$ first. We can average $g(y)$ using the distribution of Y .

EXAMPLE

Suppose that Y has the following distribution:

y	0	1	2	3
$\Pr(Y = y)$	$1/8$	$1/4$	$3/8$	$1/4$

Then

$$E(Y) = 0 \cdot \frac{1}{8} + 1 \cdot \frac{1}{4} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{1}{4} = \frac{7}{4}.$$

What is $E(Y^2)$?

$$E(Y^2) = 0^2 \cdot \frac{1}{8} + 1^2 \cdot \frac{1}{4} + 2^2 \cdot \frac{3}{8} + 3^2 \cdot \frac{1}{4} = 4.$$

Notice that $E(Y^2) \neq E(Y)^2$: here $4 \neq 49/16$.

A CONTINUOUS EXAMPLE

Suppose that Y has density

$$f(y) = \begin{cases} y/2 & \text{if } 0 \leq y \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

First, it is in fact a density:

$$\int_{-\infty}^{+\infty} f(y) dy = \int_0^2 \frac{y}{2} dy = 1.$$

Now,

$$E(Y^2) = \int_0^2 y^2 \frac{y}{2} dy = \left[\frac{y^4}{8} \right]_0^2 = 2.$$

LINEARITY OF EXPECTATION

THEOREM

If X, Y are random variables and a, b are constants, then

$$\mathbf{E}(aX + bY) = aE(X) + bE(Y).$$

In particular,

- $E(c) = c$ if c is constant,
- $E(aX) = aE(X)$,
- $E(X_1 + \cdots + X_n) = E(X_1) + \cdots + E(X_n)$.

This does *not* require independence. This is one reason why expectation is so useful.

LINEARITY: DISCRETE CASE

Suppose that X, Y are discrete. Then

$$\begin{aligned} E(aX + bY) &= \sum_x \sum_y (ax + by) \Pr(X = x, Y = y) \\ &= a \sum_x x \left(\sum_y \Pr(X = x, Y = y) \right) + b \sum_y y \left(\sum_x \Pr(X = x, Y = y) \right) \\ &= a \sum_x x \Pr(X = x) + b \sum_y y \Pr(Y = y) \\ &= aE(X) + bE(Y). \end{aligned}$$

Again: nowhere did we use independence.

PRODUCTS

Linearity tells us what happens to sums. It does not tell us what happens to products.

In general,

$$E(XY) \neq E(X)E(Y).$$

But if X and Y are independent, then

$$E(XY) = E(X)E(Y).$$

QUESTION 1

Let X be Bernoulli(1/2) and let $Y = X$. Are X and Y independent? Calculate $E(XY)$ and $E(X)E(Y)$.

BERNOULLI RANDOM VARIABLES

Suppose that X is Bernoulli(p). Then X is 1 with probability p and 0 with probability $1 - p$.

Therefore,

$$E(X) = 1 \cdot p + 0 \cdot (1 - p) = p.$$

There is a useful way to write any event as a random variable. Given an event A , let

$$\mathbb{1}_A = \begin{cases} 1 & \text{if } A \text{ occurs,} \\ 0 & \text{otherwise.} \end{cases}$$

Then $\mathbb{1}_A$ is Bernoulli($\Pr(A)$), so

$$E(\mathbb{1}_A) = \Pr(A).$$

EXPECTATION OF A BINOMIAL RANDOM VARIABLE

Suppose that $X_1, \dots, X_n$ are independent Bernoulli(p) random variables and

$$S = X_1 + \dots + X_n.$$

Then S is Binomial(n, p).

By linearity,

$$\begin{aligned} \mathbb{E}(S) &= \mathbb{E}(X_1 + \dots + X_n) \\ &= \mathbb{E}(X_1) + \dots + \mathbb{E}(X_n) \\ &= \underbrace{p + \dots + p}_{n \text{ times}} = np. \end{aligned}$$

Much easier than applying the binomial probability formula and adding.

VARIANCE

Expectation tells us where a distribution is centered. But it doesn't tell us how spread out it is.

Could we use the average deviation from the mean? Let $\mu = E(Y)$. Then

$$E(Y - \mu) = E(Y) - \mu = 0.$$

Positive and negative deviations cancel. Instead, we square them.

The **variance** of Y is

$$\mathbf{Var}(Y) \equiv E[(Y - E(Y))^2] = E[(Y - \mu)^2].$$

The **standard deviation** is $\sqrt{\mathbf{Var}(Y)}$. It is measured in the same units as Y .

DISCRETE AND CONTINUOUS VARIANCE

If Y is discrete and $\mu = E(Y)$, then

$$\text{Var}(Y) = \sum_y (y - \mu)^2 \Pr(Y = y).$$

If Y is continuous with density f , then

$$\text{Var}(Y) = \int_{-\infty}^{+\infty} (y - \mu)^2 f(y) dy.$$

Variance is always nonnegative. It is zero exactly when Y is constant with probability one.

A bigger variance means that values tend to be farther away from the mean.

A USEFUL FORMULA FOR VARIANCE

THEOREM

$$\text{Var}(\mathbf{Y}) = \text{E}(Y^2) - \text{E}(Y)^2.$$

Why? Let $\mu = \text{E}(Y)$. Then

$$\begin{aligned}\text{Var}(Y) &= \text{E}[(Y - \mu)^2] \\ &= \text{E}(Y^2 - 2\mu Y + \mu^2) \\ &= \text{E}(Y^2) - 2\mu \text{E}(Y) + \mu^2 \\ &= \text{E}(Y^2) - \mu^2.\end{aligned}$$

EXAMPLE

Recall the distribution

y	0	1	2	3
$\Pr(Y = y)$	1/8	1/4	3/8	1/4.

We calculated $E(Y) = 7/4$ and $E(Y^2) = 4$. Hence

$$\text{Var}(Y) = E(Y^2) - E(Y)^2 = 4 - \left(\frac{7}{4}\right)^2 = \frac{15}{16} = 0.9375.$$

QUESTION 2

Calculate the variance and standard deviation of a fair six-sided die. You may use $E(Y) = 7/2$ and $E(Y^2) = 91/6$.

VARIANCE OF A FAIR DIE

For a fair six-sided die,

$$E(Y) = \frac{7}{2}$$

and

$$E(Y^2) = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6}.$$

Therefore,

$$\text{Var}(Y) = E(Y^2) - E(Y)^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12} \approx 2.917.$$

The standard deviation is $\sqrt{35/12} \approx 1.708$.

VARIANCE AND AFFINE TRANSFORMATIONS

Suppose that a, b are constants. Then

$$\begin{aligned} \mathbf{E}(aY + b) &= a\mathbf{E}(Y) + b, \\ \mathbf{Var}(aY + b) &= a^2\mathbf{Var}(Y). \end{aligned}$$

Why doesn't b affect the variance? Adding b moves every possible value and the mean by the same amount:

$$\begin{aligned} \mathbf{Var}(aY + b) &= \mathbf{E}\left[(aY + b - \mathbf{E}(aY + b))^2\right] \\ &= \mathbf{E}\left[(aY + b - a\mathbf{E}(Y) - b)^2\right] \\ &= a^2\mathbf{E}\left[(Y - \mathbf{E}(Y))^2\right] = a^2\mathbf{Var}(Y). \end{aligned}$$

Multiplying Y by a multiplies distances by $|a|$, so it multiplies squared distances by a^2 .

VARIANCE OF A SUM

For any two random variables,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y),$$

where

$$\text{Cov}(X, Y) \equiv \text{E}[(X - \text{E}(X))(Y - \text{E}(Y))].$$

Covariance tells us whether X and Y tend to move together.

If X and Y are independent, then $\text{Cov}(X, Y) = 0$, so

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Unlike linearity of expectation, additivity of variance *does* require a restriction such as independence.

EXERCISE

QUESTION 3

Let X be Bernoulli(p) and set $Y = X$. Calculate $\text{Var}(X + Y)$ and compare it with $\text{Var}(X) + \text{Var}(Y)$.

Why are they different?

DEPENDENCE MATTERS

Since $Y = X$,

$$X + Y = 2X.$$

Therefore,

$$\text{Var}(X + Y) = \text{Var}(2X) = 4\text{Var}(X) = 4p(1 - p).$$

On the other hand,

$$\text{Var}(X) + \text{Var}(Y) = 2p(1 - p).$$

The two variables move together perfectly, so their covariance is positive:

$$\text{Cov}(X, Y) = \text{Cov}(X, X) = \text{Var}(X) = p(1 - p).$$

Including the covariance term gives the correct answer.

BERNOULLI VARIANCE

If X is Bernoulli(p), then $X^2 = X$: both are zero when $X = 0$ and one when $X = 1$.

Therefore,

$$\begin{aligned}\text{Var}(X) &= \text{E}(X^2) - \text{E}(X)^2 \\ &= \text{E}(X) - \text{E}(X)^2 \\ &= p - p^2 = p(1 - p).\end{aligned}$$

This is largest at $p = 1/2$. A fair coin is more uncertain than a coin that almost always lands on the same side.

BINOMIAL VARIANCE

Suppose that $X_1, \dots, X_n$ are independent Bernoulli(p) random variables and $S = X_1 + \dots + X_n$.

We already know that $E(S) = np$.

Because the variables are independent,

$$\begin{aligned}\text{Var}(S) &= \text{Var}(X_1 + \dots + X_n) \\ &= \text{Var}(X_1) + \dots + \text{Var}(X_n) \\ &= \underbrace{p(1-p) + \dots + p(1-p)}_{n \text{ times}} \\ &= np(1-p).\end{aligned}$$

Thus, if S is Binomial(n, p), then

$$E(S) = np, \quad \text{Var}(S) = np(1-p).$$

THE SAMPLE MEAN

Suppose that $X_1, \dots, X_n$ are independent Bernoulli(p) random variables. The sample mean is

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}.$$

Its expectation is

$$\mathbb{E}(\bar{X}) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) = p.$$

Its variance is

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{p(1-p)}{n}.$$

Its standard deviation is $\sqrt{p(1-p)/n}$. As n grows, the sample mean becomes more tightly concentrated around p .

EXAMPLE: A POLL

Suppose that 51% of likely voters support a candidate. We randomly survey $n = 100$ voters with replacement.

If X_i indicates whether respondent i supports the candidate, then

$$\bar{X} = \frac{1}{100} \sum_{i=1}^{100} X_i$$

is the proportion of respondents who support the candidate.

We have

$$E(\bar{X}) = 0.51$$

and

$$\sqrt{\text{Var}(\bar{X})} = \sqrt{\frac{0.51 \cdot 0.49}{100}} \approx 0.05.$$

The sampling uncertainty shrinks at rate $1/\sqrt{n}$, not $1/n$.

EXERCISE

QUESTION 4

In the polling example, how large must the sample be to cut the standard deviation of the sample proportion in half?

SAMPLE SIZE AND PRECISION

The standard deviation of the sample proportion is

$$\sqrt{\frac{p(1-p)}{n}}.$$

To cut it in half, we need

$$\sqrt{\frac{p(1-p)}{n'}} = \frac{1}{2} \sqrt{\frac{p(1-p)}{n}}.$$

Squaring both sides gives $n' = 4n$.

Thus, to cut the standard deviation in half, we must quadruple the sample size: from 100 to 400 voters.

UNIFORM DISTRIBUTION

Suppose that $Y \sim \mathcal{U}[0, 1]$. Its density is $f(y) = 1$ for $y \in [0, 1]$. Then

$$\mathbb{E}(Y) = \int_0^1 y \, dy = \frac{1}{2},$$

and

$$\mathbb{E}(Y^2) = \int_0^1 y^2 \, dy = \frac{1}{3}.$$

Therefore,

$$\text{Var}(Y) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}.$$

QUESTION 5

If $X \sim \mathcal{U}[a, b]$, write X as $(b - a)Y + a$ for $Y \sim \mathcal{U}[0, 1]$. Use the rules for affine transformations to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

EXERCISE

QUESTION 6

Let Y_1, Y_2 be independent draws from the same distribution as Y . Calculate the expected squared distance

$$\mathbb{E}[(Y_1 - Y_2)^2]$$

in terms of $\text{Var}(Y)$.

AVERAGE SQUARED DISTANCE

Since Y_1, Y_2 are independent and have the same distribution as Y ,

$$\begin{aligned}\mathbb{E}[(Y_1 - Y_2)^2] &= \mathbb{E}(Y_1^2) - 2\mathbb{E}(Y_1 Y_2) + \mathbb{E}(Y_2^2) \\ &= \mathbb{E}(Y^2) - 2\mathbb{E}(Y_1)\mathbb{E}(Y_2) + \mathbb{E}(Y^2) \\ &= 2\mathbb{E}(Y^2) - 2\mathbb{E}(Y)^2 \\ &= 2\text{Var}(Y).\end{aligned}$$

Thus, variance has a second interpretation: it is half the expected squared distance between two independent draws.

POTENTIAL OUTCOMES

We can use random variables and expectations to define causal effects.

Let D_i indicate whether unit i receives a treatment:

$$D_i = \begin{cases} 1 & \text{if unit } i \text{ receives the treatment,} \\ 0 & \text{otherwise.} \end{cases}$$

Let Y_i be the observed outcome of interest.

For each unit i , we define two **potential outcomes**:

- Y_{1i} : the outcome that unit i would have under treatment,
- Y_{0i} : the outcome that unit i would have without treatment.

These are called “potential” because only one of them is realized.

TREATMENT EFFECTS

The **treatment effect** for unit i is

$$\tau_i = Y_{1i} - Y_{0i}.$$

The observed outcome is

$$Y_i = Y_{D_i i} = D_i Y_{1i} + (1 - D_i) Y_{0i}.$$

In other words,

$$Y_i = \begin{cases} Y_{1i} & \text{if } D_i = 1, \\ Y_{0i} & \text{if } D_i = 0. \end{cases}$$

We observe Y_{1i} for treated units and Y_{0i} for untreated units. We never observe both for the same unit.

THE FUNDAMENTAL PROBLEM OF CAUSAL INFERENCE

| We can never observe both Y_{1i} and Y_{0i} for the same unit i .

Therefore, we can never directly observe the individual treatment effect

$$\tau_i = Y_{1i} - Y_{0i}.$$

If unit i receives treatment, we see Y_{1i} but not Y_{0i} . If unit i doesn't receive treatment, we see Y_{0i} but not Y_{1i} .

This is the **fundamental problem of causal inference**. We need assumptions that allow us to learn about the missing potential outcomes from other units.

AVERAGE TREATMENT EFFECT

Individual treatment effects are unobservable. We usually focus on averages.

The **average treatment effect** is

$$\tau_{ATE} = \frac{1}{N} \sum_{i=1}^N (Y_{1i} - Y_{0i}) = E(Y_{1i} - Y_{0i}).$$

By linearity of expectation,

$$\tau_{ATE} = E(Y_{1i}) - E(Y_{0i}).$$

This is still not identified from the observed outcomes alone: for every unit, one of the two potential outcomes is missing.

The rest of causal inference is largely about assumptions under which we can identify this quantity.

OTHER CAUSAL QUANTITIES

The **average treatment effect on the treated** is

$$\tau_{ATT} = E(Y_{1i} - Y_{0i} \mid D_i = 1).$$

If $N_1 = \sum_{i=1}^N D_i$ is the number of treated units, then

$$\tau_{ATT} = \frac{1}{N_1} \sum_{i=1}^N D_i (Y_{1i} - Y_{0i}).$$

The **conditional average treatment effect** for units with characteristics $X_i = x$ is

$$\tau_{CATE}(x) = E(Y_{1i} - Y_{0i} \mid X_i = x).$$

Here X_i should be measured before the treatment.

QUESTION 7

Define the average treatment effect on the untreated, τ_{ATC} .

ATE VERSUS ATT

Why might τ_{ATT} differ from τ_{ATE} ?

Suppose that people choose whether to receive a job-training program. The people who choose it may be precisely those who benefit most from it.

Then treatment effects and treatment status are associated:

$$E(Y_{1i} - Y_{0i} \mid D_i = 1) \neq E(Y_{1i} - Y_{0i}).$$

Hence,

$$\tau_{ATT} \neq \tau_{ATE}.$$

ATE and ATT answer different questions. ATE asks what would happen if we treated a randomly chosen unit. ATT asks about the units who actually received treatment.

ILLUSTRATION

Suppose we observe four units:

i	D_i	Y_i	Y_{1i}	Y_{0i}	τ_i
1	1	3	3	?	?
2	1	1	1	?	?
3	0	0	?	0	?
4	0	1	?	1	?

The observed difference in means is

$$\tilde{\tau} = E(Y_i \mid D_i = 1) - E(Y_i \mid D_i = 0) = \frac{3+1}{2} - \frac{0+1}{2} = 1.5.$$

Is this the ATE? We don't know. We need the four missing potential outcomes.

EXERCISE

Suppose, hypothetically, that the full table is

i	D_i	Y_i	Y_{1i}	Y_{0i}	τ_i
1	1	3	3	0	3
2	1	1	1	1	0
3	0	0	1	0	1
4	0	1	1	1	0

QUESTION 8

Calculate the observed difference in means, the ATE, and the ATT.

ILLUSTRATION: AVERAGE TREATMENT EFFECT

Suppose, hypothetically, that the full table is

i	D_i	Y_i	Y_{1i}	Y_{0i}	τ_i
1	1	3	3	0	3
2	1	1	1	1	0
3	0	0	1	0	1
4	0	1	1	1	0

Then

$$\tau_{ATE} = E(\tau_i) = \frac{3 + 0 + 1 + 0}{4} = 1.$$

But the observed difference in means was $\tilde{\tau} = 1.5$.

Why are they different? The treated and untreated groups differ in more than their treatment status: the missing potential outcomes matter.

ILLUSTRATION: TREATMENT EFFECT ON THE TREATED

Using the same hypothetical potential outcomes,

i	D_i	Y_i	Y_{1i}	Y_{0i}	τ_i
1	1	3	3	0	3
2	1	1	1	1	0
3	0	0	1	0	1
4	0	1	1	1	0

For the treated units,

$$\tau_{ATT} = E(\tau_i \mid D_i = 1) = \frac{3 + 0}{2} = 1.5.$$

Here $\tau_{ATT} \neq \tau_{ATE}$ because treatment effects differ across units and the treated units have larger effects on average.

WHAT WOULD MAKE A DIFFERENCE IN MEANS WORK?

We would like to replace the missing counterfactual means with observed means from the other group.

For the ATE, it would be enough to have

$$E(Y_{1i} \mid D_i = 1) = E(Y_{1i} \mid D_i = 0)$$

and

$$E(Y_{0i} \mid D_i = 1) = E(Y_{0i} \mid D_i = 0).$$

In words: treated and untreated units would have the same average potential outcomes under either treatment status.

Random assignment makes these equalities plausible. Then the observed difference in means identifies the ATE.

